

Problem 3

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Problem 1 Suppose that $f(5) = 7$, $f'(5) = 8$, $g(5) = 3$, and $g'(5) = -4$. Find:

- (a) $(fg)'(5)$.
- (b) $\left[\frac{d}{dx} \left(\frac{f}{g} \right) \right]_{x=5}$
- (c) $\left(\frac{g}{f} \right)'(5)$
- (d) $\left[\frac{d}{dx} \left(\frac{g(x)}{x+2} \right) \right]_{x=5}$
- (e) $\left[\frac{d}{dx} \left(\frac{xf(x)}{g(x)} \right) \right]_{x=5}$
- (f) $\lim_{x \rightarrow 5} \frac{x^2 g(x) - 75}{x - 5}$. **EXPLAIN.**
- (g) $\lim_{x \rightarrow 5} \frac{\frac{x+2}{f(x)} - 1}{x - 5}$. **EXPLAIN.**